INORGANIC CUMULATIVE EXAM

September 19, 1992

Polydentate ligands

Answer as many of the following as you can.

1. For each of the names or nick-names below, sketch the molecule or ion, showing all atoms and bonds (and the charge, if it is an ion).

   a) acetylacetonato, "acac"

   b) "bipy" (or "bpy")

   c) diethylenetriamine

   d) "en"

   e) "edta"

   f) oxalato

   g) tetrazacyclotetradecane ("cyclam")

   h) triethylenetetramine

   i) "Cp" (or "Cpd")

   j) "diphos"

2. The following are sub-section titles under the heading of POLYDENTATE LIGANDS in inorganic texts. Give the main idea, or an example, for each of these concepts:

   a) Flexible <u>vs</u> rigidily-planar tridentate ligands

   b) Tripod quadridentates

   c) Macrocyclic quadridentates

   d) crown ethers

   e) cryptates

   f) conformation of chelate rings

   g) optically active chelates

   h) ring size and chelation

3. No cume on chelates would be complete without asking for a definition of "the chelate effect", or the driving force for chelation. Give a chemical reaction which illustrates the chelate effect.